ShopSmart Retail Ltd.

Week 1 - Foundations of Machine Learning

AnalystLab Africa Internship Programme | Mission ML01

Executive Summary

This project builds baseline machine learning models from ShopSmart's historical retail data. The final integrated dataset contains 50,000 unique orders and 50 analytical columns before the later Age_Group feature was added. A linear regression model predicts total order value, while a logistic regression model predicts whether an order will be returned.

1. Business Problem

Regression: predict *Order_Total* to understand customer and transaction characteristics associated with spending.
Classification: predict *Return_Flag* to identify orders with higher predicted return risk.

2. Data Preparation

The source tables were loaded and merged into one master dataset. The workflow addressed missing values, duplicates, categorical encoding, numerical scaling, and train/test splitting. The final quality check reported 50,000 rows, 50,000 unique Order IDs, zero duplicate Order IDs, 2,442 returned orders, and a 4.88% return rate. The final saved split contains 40,000 training rows and 10,000 testing rows, with the 4.88% return rate preserved in each split.

3. Model Development and Evaluation

The regression model uses Linear Regression. The classification model uses Logistic Regression. Both models use preprocessing pipelines for numerical and categorical variables. The regression model achieved $R^2 = 0.8830$. The classification results are much weaker for the returned class: only 4.88% of all orders are returns, and precision for returned orders is approximately 0.05, indicating many false-positive return predictions.

Model / Metric	Result
Linear Regression - MAE	31,010.22
Linear Regression - RMSE	41,268.26
Linear Regression - R^2	0.8830
Logistic Regression - Accuracy	0.5430
Logistic Regression - Precision	0.0475
Logistic Regression - Recall	0.4385
Logistic Regression - F1	0.0856

4. Feature Importance

For order value, the largest fitted coefficients were Unit_Price (85,388.74) and Quantity (71,148.07), followed by brand indicators. This is consistent with the construction of Order_Total from quantity, unit price, discount and shipping cost, so the strong regression result should be interpreted with that relationship in mind.

For product returns, the largest absolute logistic-regression coefficients were brand indicators, including Gentry-Ward (-2.636), Allen, Lewis and Hoover (-2.595), Martin, Haley and Parrish (-2.585), and Mitchell-Mcdaniel

(-2.523). Coefficient magnitude indicates predictive influence within the fitted model; it does not establish causality.

5. Business Insights

Finding	Result
Highest average customer-segment order value	Silver: 154,578.70
Highest average age-group order value	55-64: 156,209.30
Highest total-sales store	STR015, ShopSmart Lagos (North): 317,657,047
Observed return rate	4.88%
Lowest product-level test MAE shown	Beauty Item 189: 16,456.72 (16 test orders)

6. Business Use

The regression model can support sales forecasting, customer segmentation, targeted promotions and inventory planning. The classification model can be used as an exploratory decision-support tool for identifying transactions with higher predicted return risk and investigating product, customer and operational characteristics associated with returns.

7. Recommendations

Use the regression model as a baseline rather than a final production system. For return prediction, address class imbalance and compare alternative models and decision thresholds before operational use. Investigate high-return brands and stores in more detail, and combine predictions with business rules and managerial review.

8. Limitations

Linear regression assumes a linear relationship between predictors and the target, while logistic regression assumes linearity in the log-odds. Returns are highly imbalanced. Available variables may omit relevant drivers, historical patterns may change, and some predictors may be correlated. Most importantly, Order_Total is constructed from quantity, unit price, discount and shipping cost, so using these components as predictors partly explains the high regression performance. The models identify predictive associations, not causal effects.

9. Model Persistence

The trained linear regression model, logistic regression model, scaler and encoder are saved with Joblib. The accompanying prediction script loads the saved models, accepts user input, predicts order value and reports return probability. The model files were created with scikit-learn 1.6.1, which is pinned in requirements.txt for compatibility.